3-masala yechimi.

a) Har bir sikl davomida sistema kameradan gazni sōrib olib, uni atmosferaga siqib chiqaradi. $V_0 = 0$ bōlgani uchun sōruvchi klapan porshen chapga harakatlana boshlagan paytda ochiladi. Porshen chapdagi eng chekka holatida bōlganida gazning $\Delta V/(V+\Delta V)$ qismi silindr ichida bōladi. Porshen ōngga harakatlanganidan keyin bu silindrdagi gazning hammasi tashqariga chiqarib yuboriladi. Bitta sikldan keyingi bosim quyidagicha topiladi (izotermik jarayon)

$$P_{dastl.} \cdot V = P_{keyingi} \cdot (V + \Delta V), \text{ bundan}$$

$$\boxed{P_{keyingi} = P_{dastl.} \cdot \left(\frac{V}{V+\Delta V}\right).} \quad (1\ ball)$$

$N = \frac{t}{\Delta t}$ ta sikldan keyingi bosim:

$$\boxed{P(t) = P_a \cdot \left(\frac{V}{V+\Delta V}\right)^{t/\Delta t}.} \quad (1\ ball)$$

Bu ifoda texnik jihatdan tōgri, ammo uni quyidagicha tarzda ancha soddalashtirish mumkin:

$$\boxed{P(t) = P_a\left(1+\frac{\Delta V}{V}\right)^{-t/\Delta t} = P_a\left((1+x)^{1/x}\right)^{-Rt/V},} \quad (1\ ball)$$

bu yerda $x = \Delta V/V \ll 1$.

e ning ta'rifi $e = \lim\limits_{x\to 0}(1+x)^{1/x}$ dan foydalanib,

$$\boxed{P(t) = P_a \cdot e^{-Rt/V}} \quad (1\ ball)$$ ga ega bōlamiz.

b) Porshen chapdagi eng chekka vaziyatda bōlganida bosim $P(t)$ga va temperatura T_a ga teng. Porshen ōngga harakatlanigani sari bosim P_a ga teng bōlgunicha (va chiqaruvchi klapan ochilgunicha) gaz adiabatik tarzda siqiladi.

$$\boxed{PV^{\gamma} = const, \quad \frac{PV}{T} = const} \quad (1\ ball)$$

Bu yerdagi 2-formulani γ darajaga kōtarib, 1-formulani γ darajaga kōtarilgan 2-formulaga bōlib yuborilsa,

$$\boxed{T^{\gamma} \cdot P^{1-\gamma} = const}, \text{ yoki } \boxed{T \cdot P^{\frac{1-\gamma}{\gamma}} = T \cdot P^{\left(\frac{1}{\gamma}-1\right)} = const}$$

hosil bōladi (1 ball)

$$T_a \cdot (P(t))^{\frac{1-\gamma}{\gamma}} = T_{tashq.}(t) \cdot (P_a)^{\frac{1-\gamma}{\gamma}}, \quad \gamma = \frac{5}{3} \text{ — bir atomli gaz}$$

$$T_{tashq.}(t) = T_a \cdot \left(\frac{P_a}{P(t)}\right)^{\left(1-\frac{1}{\gamma}\right)} = T_a \cdot \left(\frac{P_a}{P(t)}\right)^{2/5}.$$

$$T_{tashq.}(t) = T_a \cdot \left(1 + \frac{\Delta V}{V}\right)^{\frac{2t}{5\Delta t}} = T_a \cdot e^{\frac{2Rt}{5V}}$$

(1 ball) (1 ball)

3-masala davomi

c) $V_0 > 0$ bo'lgani uchun, porshen harakatlana boshlaganida (chapga) so'ruvchi klapan darrov ochilmaydi, aksincha, u silindrdagi bosim kameradagi bosimga teng bo'lganidagina ochiladi. Silindrning kengayishi adiabatik bo'lgani uchun, $PV^\gamma = const$, ya'ni:

$$P_a \cdot V_0^\gamma = P_1 \cdot (V_0 + \Delta V)^\gamma \Rightarrow P_1 = P_a \left(\frac{V_0}{V_0 + \Delta V}\right)^\gamma = P_a \left(1 + \frac{\Delta V}{V_0}\right)^{-\gamma}$$

(1 ball)

$$P_{min} = P_a \left(1 + \frac{\Delta V}{V_0}\right)^{-\frac{\gamma t}{\Delta t}}, \quad t \to \infty \Rightarrow P_{min} \to 0$$

(1 ball)